

Soldering Iron Tip Preserver

RAJ K. GORKHALI

Although 60/40 solders melt at about 200°C, the tip temperature of a soldering iron should be about 370°C. This is necessary to make a good joint quickly without the risk of overheating delicate components. One should not hold the tip of

the iron to the joint for too long at such high temperature.

Unfortunately, at this temperature, the tip oxidises rapidly, requiring constant cleaning. That's where this circuit can help as it keeps the soldering tip to just below 200°C while the iron is at rest. Oxidisation is then negligible and the iron can be brought back to soldering temperature in just a few seconds when needed. In addition, during normal soldering operation, where the



Test Points

Test point	Details
TP0	GND
TP1, TP2	24V AC
TP3	12V
TP4	Approx. 12V

PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
- D1 - 12V, 1W zener diode
- D1 - 1N4007 rectifier diode
- BR1 - 1A bridge rectifier module
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

- R1 - 180-ohm, 5W
- R2 - 180-kilo-ohm
- R3 - 15-kilo-ohm
- R4 - 560-kilo-ohm
- R5 - 100-kilo-ohm
- R6 - 18-kilo-ohm
- R7 - 10-kilo-ohm

Capacitors:

- C1 - 100µF, 63V electrolytic
- C2 - 220µF, 63V electrolytic
- C3 - 47µF, 63V electrolytic
- C4, C5 - 0.01µF ceramic
- C6 - 470µF, 63V electrolytic

Miscellaneous:

- RL1 - 12V, 1C/O relay
- CON1 - Three-pin connector
- CON2 - Two-pin connector

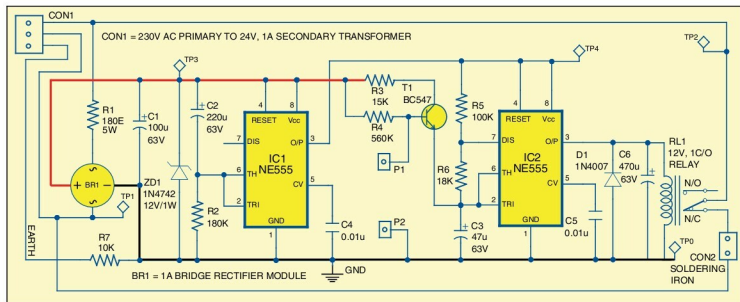


Fig. 1: Circuit of soldering iron tip preserver

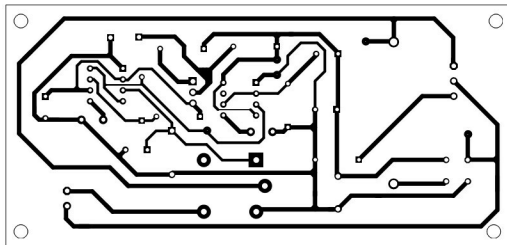


Fig. 2: An actual-size, single-side PCB for the soldering iron tip preserver

iron returns to rest only momentarily, the tip temperature is unaffected because of the thermal inertia of iron.

Circuit and working

Fig. 1 shows the circuit of soldering iron tip preserver. Two 555 timers (IC1 and IC2) are the heart of the circuit. IC1 is wired in monostable configuration and provides an initial warm-up time of about 45 seconds to bring the iron up to the required soldering temperature. At the end of this period, its output pin (3) becomes high, enabling IC2 (wired in astable